

Linear System

→ The highest power of all variables must be 1.

Example:
$$\begin{cases} x_1 + 2x_2 + x_3 = 0 \\ x_1 - 2x_2 + 2x_3 = 4 \\ 2x_1 + 12x_2 - 2x_3 = 4 \end{cases}$$
 A linear system of equation
Here, Number of unknown/variable $n = 3$

General Form:

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = b_1$$

a_{ij} = Coefficient

$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = b_2$$

b_i = Known Constant

$\vdots$

x_j = Unknown Variable

$$a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n = b_n$$

Matrix Form:

$$\underbrace{\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}}_A \underbrace{\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}}_X = \underbrace{\begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix}}_b$$

$$\Rightarrow AX = b$$

Solving Linear System:

1. Must be a square matrix. ($n \times n$)
2. Must be a non singular (Invertible)

5.1: Triangular System:

- Upper Triangular Matrix

$$\textcircled{*} \det(T) = \prod_{i=1}^n t_{ii}$$

- Lower Triangular Matrix

Solving Triangular System:

Using a lower triangular, ($n=4$)

$$AX = b$$

$$\begin{pmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix} \Rightarrow \begin{aligned} l_{11}x_1 &= b_1 \\ l_{21}x_1 + l_{22}x_2 &= b_2 \\ l_{31}x_1 + l_{32}x_2 + l_{33}x_3 &= b_3 \\ l_{41}x_1 + l_{42}x_2 + l_{43}x_3 + l_{44}x_4 &= b_4 \end{aligned}$$



$$\begin{aligned}
 x_1 &= \frac{b_1}{l_{11}} \\
 x_2 &= \frac{b_2 - l_{21}x_1}{l_{22}} \\
 x_3 &= \frac{b_3 - l_{31}x_1 - l_{32}x_2}{l_{33}} \\
 x_4 &= \frac{b_4 - l_{41}x_1 - l_{42}x_2 - l_{43}x_3}{l_{44}}
 \end{aligned}$$

This is called forward substitution
(For lower)

And, Backward substitution is for upper.

Total number of operation required is n^2 . [operation: +, -, x, /]

(*) Prove that the total number of operation required for backward substitution is n^2 .

$$\Rightarrow \text{let, } n=4, \begin{pmatrix} l_{11} & 0 & 0 & 0 \\ l_{21} & l_{22} & 0 & 0 \\ l_{31} & l_{32} & l_{33} & 0 \\ l_{41} & l_{42} & l_{43} & l_{44} \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \\ b_4 \end{pmatrix}$$

$$\text{Now, } l_{11}x_1 = b_1$$

$$x_1 = \frac{b_1}{l_{11}} \rightarrow 1 \text{ div}$$

$$l_{21}x_1 + l_{22}x_2 = b_2$$

$$x_2 = \frac{b_2 - l_{21}x_1}{l_{22}} \rightarrow 1 \text{ div, 1 mul, 1 sub}$$

$$l_{31}x_1 + l_{32}x_2 + l_{33}x_3 = b_3$$

$$l_{41}x_1 + l_{42}x_2 + l_{43}x_3 + l_{44}x_4 = b_4$$

$$x_3 = \rightarrow 1 \text{ div, 2 mul, 2 sub}$$

$$x_4 = \rightarrow 1 \text{ div, 3 mul, 3 sub}$$

⋮

$$x_n = \rightarrow 1 \text{ div, } (n-1) \text{ mul, }$$

$$\therefore \text{Total number of operation} = \sum_{j=1}^n [1 + (j-1) + (j-1)] \quad (n-1) \text{ sub}$$

$$= \sum_{j=1}^n [2j-1] = 2 \sum_{j=1}^n j - \sum_{j=1}^n 1 = 2 \cdot \frac{n(n+1)}{2} - n = n^2$$

$$= 2 \cdot \frac{n(n+1)}{2} - n = n^2 \quad (\text{Proved})$$

5.2 : Gaussial Elimination

1. Transform A into Triangular Matrix (Upper/Lower)

2. Using Backward/Forward substitution to solve

• Example: $x_1 + 2x_2 + x_3 = 0$

$x_1 - 2x_2 + 2x_3 = 4$

$2x_1 + 12x_2 - 2x_3 = 4$

Solve.

 $\Rightarrow$ Augmented matrix

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & -2 & 2 & 4 \\ 2 & 12 & -2 & 4 \end{array} \right)$$

$Ax = b \rightarrow Ux = b'$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{pmatrix}$$

Row operation

Upper or lower

We define row multiplier as,

1st row operation

$r_1 \rightarrow r'_1 = r_1$

$r_2 \rightarrow r'_2 = r_2 - m_{21}r_1$

$r_3 \rightarrow r'_3 = r_3 - m_{31}r_1$

(beyond)

Example: $x_1 + 2x_2 + x_3 = 0$ Solve $\begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix} = U$

$x_1 - 2x_2 + 2x_3 = 4$

$2x_1 + 12x_2 - 2x_3 = 4$

$\Rightarrow$

$$A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix} \quad b = \begin{pmatrix} 0 \\ 4 \\ 4 \end{pmatrix} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\det(A) = 8 \neq 0; \text{ Aug}(A) = (A:b) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 1 & -2 & 2 & 4 \\ 2 & 12 & -2 & 4 \end{array} \right)$$

1st row operation,

$$m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2, \quad m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1} = 1$$

$$r_1 \rightarrow r'_1 = r_1 = (1 \ 2 \ 1 \ 0)$$

$$r_2 \rightarrow r'_2 = r_2 - m_{21}r_1 = (1 \ -2 \ 2 \ 4) - 1(1 \ 2 \ 1 \ 0) = (0 \ -4 \ 1 \ 4)$$

$$r_3 \rightarrow r'_3 = r_3 - m_{31}r_1 = (2 \ 12 \ -2 \ 4) - 2(1 \ 2 \ 1 \ 0) = (0 \ 8 \ -4 \ 4)$$

$$\text{Aug}(A) \rightarrow \text{New Aug}(A) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & 8 & -4 & 4 \end{array} \right)$$

2nd Row operation,

$$m_{32} = \frac{a_{32}}{a_{22}} = \frac{8}{-4} = -2$$

$$r_3 \rightarrow r'_3 = r_3 - m_{32}r_2 = (0 \ 8 \ -4 \ 4) - (-2)(0 \ -4 \ 1 \ 4) = (0 \ 0 \ -2 \ 12)$$

$$\text{Aug}(A) = \left(\begin{array}{ccc|c} 1 & 2 & 1 & 0 \\ 0 & -4 & 1 & 4 \\ 0 & 0 & -2 & 12 \end{array} \right) \Rightarrow (U_x = b)$$

We have. $U = \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{pmatrix}$ $b = \begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix}$

$$\begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & 1 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix}$$

$-2x_3 = 12 \Rightarrow x_3 = -6$ $\therefore$ The solution is: $x_1 = 11, x_2 = -\frac{5}{2}, x_3 = -6$

$-4x_2 + x_3 = 4 \Rightarrow x_2 = -\frac{5}{2}$

$x_1 + 2x_2 + x_3 = 0 \Rightarrow x_1 = 11$

Example: The upward velocity of a rocket given at three different time in the following table:

Time(s)	5	8	12
Velocity(ms^{-1})	106.8	177.2	279.2

* The velocity data is approximated by a polynomial as

$$v(t) = b_1 t^2 + b_2 t + b_3 \quad \text{for } 5 \leq t \leq 12$$

a) Find the value of b_1, b_2 & b_3 using the gaussian elimination method.

b) " " velocity at $t=7$ second.

$\Rightarrow$ (a) $v(t) = b_1 t^2 + b_2 t + b_3$

$v(5) = b_1(5^2) + b_2(5) + b_3 = 106.8$

$\Rightarrow 25b_1 + 5b_2 + b_3 = 106.8 \quad \text{--- (i)}$

$v(8) \Rightarrow 64b_1 + 8b_2 + b_3 = 177.2 \quad \text{--- (ii)}$

$v(12) \Rightarrow 144b_1 + 12b_2 + b_3 = 279.2 \quad \text{--- (iii)}$

Aug(A). $\begin{pmatrix} 25 & 5 & 1 & 106.8 \\ 64 & 8 & 1 & 177.2 \\ 144 & 12 & 1 & 279.2 \end{pmatrix}$

$m_{21} = \frac{a_{21}}{a_{11}} = \frac{64}{25} = 2.56$

$m_{31} = \frac{144}{25} = 5.76$

1st row operation.

$$r_2 \rightarrow r_2' = r_2 - m_{21} r_1 = (0 \quad -4.8 \quad -1.56 \quad -96.208)$$

$$r_3 \rightarrow r_3' = r_3 - m_{31} r_1 = (0 \quad -16.8 \quad -4.76 \quad -335.968)$$

$$\text{New Aug}(A) \rightarrow \left(\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & -16.8 & -4.76 & -335.968 \end{array} \right)$$

$$m_{32} = \frac{-16.8}{-4.8} = 3.5$$

2nd Row op.

$$r_3 \rightarrow r_3' = r_3 - m_{32} r_2 = (0 \quad 0 \quad 0.7 \quad 0.76)$$

$$\text{Aug}(A) = \left(\begin{array}{ccc|c} 25 & 5 & 1 & 106.8 \\ 0 & -4.8 & -1.56 & -96.208 \\ 0 & 0 & 0.7 & 0.76 \end{array} \right) \cong (Ux = b)$$

$$0.7 b_3 = 0.76 \Rightarrow b_3 = \frac{0.76}{0.7} = \boxed{1.085}$$

$$b_2 = \boxed{19.69}$$

$$b_1 = \boxed{0.2597}$$

$$(b) \quad v(t) = 0.2597 t^2 + 19.69 t + 1.085$$

$$\therefore v(7) = \boxed{151.643}$$

5.3 : LU Decomposition

$$x_1 + 2x_2 + x_3 = 0$$

$$x_1 - 2x_2 + 2x_3 = 4$$

$$2x_1 + 12x_2 - 2x_3 = 4$$

$$\Rightarrow A = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad b = \begin{pmatrix} 0 \\ 4 \\ 4 \end{pmatrix}$$

LU Decomposition Idea:

$$A = LU$$

L = Lower triangular Matrix (with 1s on diagonal); U = Upper Triangular Matrix

$$L = \begin{pmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{pmatrix}$$

$$U = \begin{pmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{pmatrix}$$

$$\text{To solve, } \begin{cases} L y = b \\ U x = y \end{cases} \Rightarrow Ax = b$$

Example:

Step 1: Finding coefficient Matrix

$$A^{(1)} = \begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix}$$

$$m_{21} = \frac{a_{21}}{a_{11}} = \frac{1}{1} = 1; \quad m_{31} = \frac{a_{31}}{a_{11}} = \frac{2}{1} = 2;$$

Step 2: Find Frobenius Matrix $F^{(1)}$

$$F^{(1)} = \begin{pmatrix} 1 & 0 & 0 \\ -m_{21} & 1 & 0 \\ -m_{31} & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix}$$

Step 3: Find $A^{(2)} = F^{(1)} \times A^{(1)}$

$$A^{(2)} = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ -2 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 1 & -2 & 2 \\ 2 & 12 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -1 \\ 0 & 8 & -4 \end{pmatrix}$$

$$m_{32} = \frac{a_{32}}{a_{22}} = \frac{8}{-4} = -2$$

Step 4: Find Frobenius Matrix, $F^{(2)}$

$$F^{(2)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -m_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$$

Step 5: Find $A^{(3)} = F^{(2)} \times A^{(2)}$

$$A^{(3)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -1 \\ 0 & 8 & -4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -1 \\ 0 & 0 & -2 \end{pmatrix} = U$$

Step 6: Find L

$$L = \begin{pmatrix} 1 & 0 & 0 \\ m_{21} & 1 & 0 \\ m_{31} & m_{32} & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix}$$

Step 7: $Ly = b$

$$\Rightarrow \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 2 & -2 & 1 \end{pmatrix} \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 4 \end{pmatrix} \Rightarrow y_1 = 0, y_2 = 4, y_3 = 12$$

Step 8: Compute $Ux = y$

$$\Rightarrow \begin{pmatrix} 1 & 2 & 1 \\ 0 & -4 & -1 \\ 0 & 0 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 4 \\ 12 \end{pmatrix} \Rightarrow x_1 = 11, x_2 = -1.5, x_3 = -6$$

(Solutions)

Q Example: $2x + 3y + z = 9$

Solve.

$$4x + 7y + 5z = 23$$

$$-2x + 4y + 5z = 9$$

$$\Rightarrow A^{(1)} = \begin{pmatrix} 2 & 3 & 1 \\ 4 & 7 & 5 \\ -2 & 4 & 5 \end{pmatrix}$$

$$m_{21} = \frac{4}{2} = 2, \quad m_{31} = -1$$

$$F^{(1)} = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$$

$$A^{(2)} = F^{(1)} A^{(1)} = \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 7 & 6 \end{pmatrix}$$

$$m_{32} = 7$$

$$F^{(2)} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -7 & 1 \end{pmatrix}$$

$$A^{(3)} = F^{(2)} A^{(2)}$$

$$= \begin{pmatrix} 2 & 3 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & -15 \end{pmatrix} = U \quad ; \quad L = \begin{pmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 7 & 1 \end{pmatrix}$$

$$Ly = b$$

$$y_1 = 9, \quad y_2 = 5, \quad y_3 = -17$$

$$Ux = y$$

$$x = \frac{23}{15}, \quad y = \frac{8}{5}, \quad z = \frac{17}{15}$$